

LEARNER FILE DIVIDER

CONTENT	CHECKED
LEARNER WORKBOOK	
SUMMATIVE ASSESSMENT	

KNOWLEDGE MODULES

KM ID	TITLE	Credits
KM – 09	Fault Finding, NQF Level 4.	3

30
 30
 100%

Module #	671101000-KM-9:
NQF Level	level 4,
Notional hours	30
Credit(s)	Cr 3
Occupation or Specialisation	671101000: Occupational Certificate: Electrician
Associated Occupation	671101: Electrician

FAULT FINDING

SUMMATIVE ASSESMENT

Details	Please Complete this Section
Name	Xolani Collen Malinga
Contact Address	32467 Mandela Str
Telephone (H)	0747732317
Telephone (W)	0747732317
Facsimile	N/A
Cellular	0747732317
E-mail	collenx14@gmail.com

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Welcome to your Summative Assessment!

This document deals with the summative assessment of your competence towards the outcomes which you need to achieve. “Summative Assessment is assessment for making a judgement about achievement. This is carried out when a learner is ready to be assessed at the end of a programme of learning” (SAQA: Criteria and Guidelines for Assessment Policy Document,).

LEARNER ASSESSMENT PLAN

Unit code	TITLES	NQF level	Credits	✓

Activity	Evidence of activity will be found where	Place & planned date of activity	Date Completed
Training	Classroom training registers	Training Provider Date:	
Self assessment	Assessment contract signed & dated	Learner file Date:	
Assessment contract	Assessment contract signed & dated	Learner file Date:	
Initial meeting	Assessor briefing checklist	Learner file Date:	
	Assessment contract & assessment plan	Learner file Date:	
Formative Assessment	Assessment instruments	Learner file Date:	
Summative Assessment	Assessment instruments	Learner file Date:	
Other Evidence	Research portfolio (if applicable)	Learner file Date:	
Feedback	Feedback Report	Learner file Date:	
Moderation	Moderators report	Learner file Date:	
Judgement	Assessor Summary Report / Moderator report	Learner file Date:	
1 st Reassessment	Assessors summary report / instruments	Learner file Date:	
2 nd Reassessment	Assessors summary report / instruments	Learner file Date:	

Special arrangements for assessment

Place
Language
Resources
Barriers

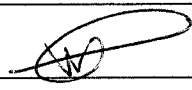
People to be involved with assessment

Learner: <u>Xolani Malinga</u>	Manager:
Trainer:	Mentor / Coach:
Assessor:	Moderator:

Next steps for learning

Resources required for this assessment

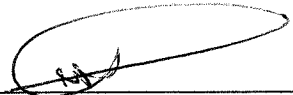
Guidelines to the learner:

<u>Xolani Malinga</u> 

Learners Name:

Xolani Collen Malinga

Learner's signature:



Date: _____

Assessors Name: _____

Assessor's signature: _____

Date: _____

ASSESSMENT APPEALS PROCEDURE

1. A learner has the right to appeal under the following circumstances
 - If the laid down assessment procedures were not followed during assessments
 - If not all evidence available was taken into account during the assessment
 - The assessor was not a subject matter expert or did not have a subject matter expert during the assessment process
 - The assessor did not assess according to the performance criteria and range statement stipulated in the unit standard
 - Not all the range items were available for assessment
2. A learner bringing an appeal against a decision of the assessment will lodge such an appeal with the assessor and the internal moderator within 2 days of the assessment feedback session.
3. A learner bringing an appeal should complete the "Learner's Notice of Assessment Appeal" form before the Appeal Hearing. The form should be handed to the internal moderator or a representative of the SETA.
4. Should the internal moderator re-affirm the assessor's decision, the learner may appeal to the external verifier within 2 days after the initial moderator's feedback session. The external verifier's decision will be final. Should the external verifier re-affirm the assessors' decision, the cost for re-evaluation will be upon the learner. Should the verifier's decision differ from the assessor's decision, the cost for re-evaluation will be borne by the assessor.

ASSESSMENT APPEAL APPLICATION FORM

LEARNER'S NOTICE OF ASSESSMENT APPEAL

TO: The Internal Moderator

A meeting with the internal moderator is hereby requested to discuss the outcome of my assessment.

Internal moderator name _____

Date of submission: _____

Name of employee assessed: _____

Name of Assessor: _____

Date of feedback session: _____

Grounds for Appeal

No	Tick the applicable ground(s) for appeal	Tick
1	The assessment did not follow the laid down procedure	
2	Not all evidence available was taken into account during the assessment	
3	The assessor was not a subject matter expert nor was a subject matter expert present during the assessment process	
4	The assessment was not according to the performance criteria and the range statement stipulated in the unit standard	
5	Not all the range items were available for the assessment	

Reasons for Appeal

No	Please give detailed reasons for the choice(s) above
1	
2	
3	
4	
5	

Learner's signature: _____

Date: _____

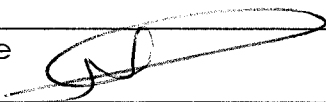
Employee witness: _____

Date: _____

PRE-ASSESSMENT MEETING CHECKLIST

Points Assessor must cover in the initial meeting with the learner - Please tick

Item	Points to be covered	Tick
1	Welcome the candidate and put them at ease	✓
2	Explain the purpose of the meeting (why you are there and how long the meeting will take)	✓
3	Explain the ▪ NQF ▪ Credits ▪ Certification process ▪ Learning pathways	✓
4	Explain ▪ Who is involved in the assessment and their role (learners, coach, assessors, managers, moderators) ▪ Principles of assessment (fairness, confidentiality, validity, sufficiency)	✓
5	Explain the assessment process? ▪ Check learner readiness for assessment (logbook / self assessment) ▪ Assessment contract to be completed ▪ Preparation of learner (this meeting) ▪ The assessment (observation and knowledge questionnaire) ▪ Judgement of the evidence ▪ Outcome of assessment (competent, not yet competent, need further evidence)	✓
6	Give Learner copies of the following documentation and explain each document ▪ The Assessment Guide which includes ○ The relevant unit standard (s) ○ Assessment contract ○ Assessment plan ○ Observation checklist ○ Knowledge checklist	✓
7	Discuss the assessment plan (complete the assessment plan document) ▪ Allow the learner to participate in the decisions made ▪ Agree on dates, time and venue for the assessment and feedback ▪ Agree on evidence the learner can submit ▪ Agree and explain the assessment methods ▪ Identify and discuss special assessment needs of the candidate ▪ Identify and eliminate unfair barriers (language, disability etc) ▪ Discuss and agree on witness requirements	✓
8	Tell the candidate his/her rights and responsibilities, the assessment procedures and policies ▪ How many times they may be assessed ▪ Appeals process / procedure ▪ Reassessment policy	✓
9	Ensure the assessment environment is appropriate or make special arrangements	✓
10	Discuss moderation	✓
11	Allow the learner opportunity to clarify any items discussed	✓

Learner declaration of acceptance of assessment instruments and relevant documentation: Date:	
Learners Name: <i>Xolani Colleen Malingee</i>	Signature: 
Assessors Name:	Signature:

INSTRUCTIONS:

The following sections will test your knowledge and you are required to answer all the questions.

All true and false items must be answered.

Once you have finished you may obtain feedback from your assessor on the accuracy of your answer.

LEARNING UNIT 1:

KM-09-KT01: Fault finding (100%)

Learning Outcome 1:	KT0101 Faultfinding principles for electrical circuits
Learning Outcome 2:	KT0102 Faultfinding techniques for electrical circuits
Learning Outcome 3:	KT0103 Safety during faultfinding

Task	Questions Description	Marks
1	Explain fault finding principles and techniques for alternating and direct current main supply circuits	20

Fault Finding Principle

Fault finding is a systematic process used to identify and correct problems in electrical circuits

Understand the Circuits: Study circuit diagrams and know how the system should operate under normal conditions

Follow isolate the power before testing where possible.

Follow a logical step by step approach instead of guessing
Start from the supply and move towards the load

Use instruments like a multimeter, test lamp or Clamp meter correctly

Break the circuit into sections and test each part to locate the fault

Observe what is wrong (e.g. no power, intermittent operation, overheating)

Check Voltage, current and resistance against expected values.

Check for ~~burnt~~ components, loose connections, damaged insulation or corrosion.

look for tripped ~~breakers~~ or blown fuses

Measure voltage ~~different~~ points

Helps identify ~~where~~ power is lost in the circuit

Used when the circuit ~~is~~ off

Checks if conductors ~~or~~ components are open or broken

Detects faulty ~~components~~ like resistors or coils

High or infinite resistance may indicate an / Open circuit.

Replace suspected faulty components with known good ones

Divide the circuit ~~into~~ halves and test each section to quickly locate the fault

Check phase and ~~neutral~~ supply

look for unbalanced loads in three-phase systems.

Check polarity (positive ~~and~~ negative connections)

Ensure proper battery or DC supply Voltage.

Always isolate power before working

Use lockout / tagout procedures

Avoid working on live circuits unless necessary

Use insulated tools and wear PPE

Be aware of shock and arc hazards

Inspect switches, fuses and control devices

Task	Questions Description	Marks
2	Explain Safety consideration during fault finding	10

Safety Considerations during Fault finding
lockout/Tagout Procedures.

Crucial for ensuring machines are fully shut off, de-energized and cannot be accidentally restarted while being worked on.

Electrical Isolation and Testing

Before inspection, disconnect power and use a calibrated Voltage detector to prove the circuit is dead.

Personal Protective Equipment

Always wear appropriate gear, including safety glasses, insulated gloves and safety boots, to protect against potential arc flash or chemical exposure.

Hazardous Environment Awareness

Identify potential risks beyond electricity such as hazardous fumes, or stored energy.

Use of Correct Tools

Utilize insulated tools designed for the specific Voltage and task, ensuring they are regulatory inspected and calibrated.

Risk Assessment and Supervision

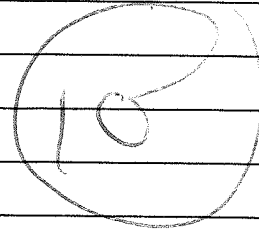
For complex, high-risk or energized testing a formal risk assessment and specialized authorization must be completed.

1. Isolate the power Supply.

~~Always~~ Switch off and lock out the Circuit before working

2. Use Insulated tools.

~~Test~~ before touching



Use a tester to Confirm the Circuit is de-energized

~~Only~~ test live Circuits when absolutely necessary and with caution.

keep the Work area dry and Clean

Water Increases risk of Electrical Shock

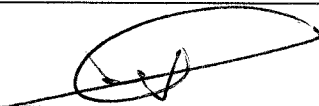
Prevents accidental electric shock

Assessor's Report

Candidate's Name	<i>Xolani Collen Malinga</i>	ID No.	951155429085
Assessor's Name		Reg. No.	
Unit Standard Title			
ASSESSMENT DECISION			
Outcome	C	NYC	Comments
OVERALL COMPETENCE JUDGEMENT:			
Comments			
Date			
Signature of Assessor	Signature of Candidate		

Candidate Feedback Report

Candidate's Name	<i>Xolani Collen Malinga</i>	ID No.	951155429085
Assessor's Name		Reg. No.	
Title			

ASSESSMENT DECISION			
Source of Evidence	C	NYC	Comment
Knowledge questions	✓		Good
Product /Work sample	✓		Good
Project	✓		Good
Indirect Evidence	✓		Good
Overall Assessment Decision			
Additional Notes			
Date			
			
Signature of Assessor			Signature of Candidate

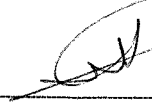
Candidate Appeal Form

Candidate's Name:	ID No.	
Assessor's Name:	Reg. No.	
Title:		
Date:		
SECTION 1		
Candidate's reason for disagreeing with the assessment decision		
Assessor's rationale for the assessment decision		
Candidate's signature		
Assessor's signature		
SECTION 2		
Moderator's decision		
Moderator's signature		

Moderator's Report

Moderator's Name		Reg. No.	
Assessor's Name		Reg. No.	
Candidate's Name	Xolani Colleen Malinga	ID No.	9511155424085.
Title	Fault Finding		
MODERATION DECISION			
Outcome	C	NYC	Comments
Overall Moderation Decision			
Feedback to Assessor			
Action Required			
Date of Moderation			
Signature of Moderator			
Signature of Assessor			
Signature of Candidate			

ASSESSMENT REVIEW

NAME of LEARNER <i>Xolani Malinga</i>		NAME of ASSESSOR	
VENUE <i>Easec Technocentric</i>		DATE of REVIEW	
Review Dimension	ASSESSOR	LEARNER/ CANDIDATE	ACTION
The principles/criteria for good assessment were achieved?	☐ Agree ☐ Disagree	☒ Agree ☐ Disagree	
The assessment related to the registered unit standard?	☐ Agree ☐ Disagree	☒ Agree ☐ Disagree	
The assessment was practical?	☐ Agree ☐ Disagree	☒ Agree ☐ Disagree	
It was time efficient and cost-effective and did not interfere with my normal responsibilities?	☐ Agree ☐ Disagree	☒ Agree ☐ Disagree	
The assessment instruments were fair, clear and understandable	☐ Agree ☐ Disagree	☒ Agree ☐ Disagree	
The assessment judgements was made against set requirements	☐ Agree ☐ Disagree	☒ Agree ☐ Disagree	
The venue and equipment was functional?	☐ Agree ☐ Disagree	☒ Agree ☐ Disagree	
Special needs were identified and the assessment plan was adjusted	☐ Agree ☐ Disagree	☒ Agree ☐ Disagree	
Feedback was constructive against the evidence required	☐ Agree ☐ Disagree	☒ Agree ☐ Disagree	
An opportunity to appeal was given	☐ Agree ☐ Disagree	☒ Agree ☐ Disagree	
The evidence was recorded	☐ Agree ☐ Disagree	☒ Agree ☐ Disagree	
LEARNER'S DECLARATION OF UNDERSTANDING			
I am aware of the moderation process and understand that the moderator could declare the assessment decision invalid			
			
Learner	Date	Assessor	Date
		Moderator	Date

9
9
100%

Module #	671101000-KM-9:
NQF Level	level 4,
Notional hours	30
Credit(s)	Cr 3
Occupation or Specialisation	671101000: Occupational Certificate: Electrician
Associated Occupation	671101: Electrician

FAULT FINDING

FORMATIVE ASSESSMENT

EARNING UNIT 1:

T01: Fault finding (100%)

- Faultfinding principles for electrical circuits
- 2 Faultfinding techniques for electrical circuits
- Safety during faultfinding

Questions Description	Marks
nce of fault finding	4

ant in electrical Systems because
correct problems quickly and safely
equipment operates efficiently
damage.

improves safety. Electrical faults
or earth leakages can cause
or injury.

only help protect people and

reduce downtime. When a fault
and repaired.

tems can return to normal
rich is important in homes
business

for maintaining safety, reducing
damage, improving efficiency and
electrical systems.

ate this Section

dinga
r Ex12B.

Task	Questions Description	Marks
2	Explain fault finding principles and techniques for alternating and direct current control circuits	5

a) Always isolate the power supply before working on the circuit

Use proper PPE and test for dead circuits before touching conductors

b) Study the circuit diagram and know how the system should operate

Identify components such as switches, relays contactors and power supplies

c) follow a step-by-step method rather than guessing. Start from the power source and move towards the load.

d) Break the circuit into sections and test each part to locate the fault faster.

e) loose connections, blown fuses, tripped breakers or faulty switches are common issues.

Fault finding techniques:

a) Check for burnt components, loose wires, broken connections or signs of overheating

b) Use a multimeter to check if Voltage is present at different points in the circuit

In AC circuits → check phase and neutral

c) follow the flow of Voltage from the supply through each component until the fault is found.

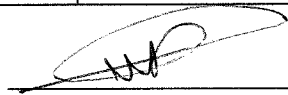
e) Replace a suspected faulty component with a known working one to confirm the fault.

SELF-ASSESSMENT

The learner must make use of the following self-evaluation checklist to rate himself against the learning outcomes of this particular training module in establishing the level of mastery of the information.

1. Not able to comply
2. Reasonable compliance (Not acceptable for final evaluation)
3. Able to comply fully

OUTCOME		1	2	3
1	KT0101 Faultfinding principles for electrical circuits			✓
2	KT0102 Faultfinding techniques for electrical circuits			✓
3	KT0103 Safety during faultfinding			✓



Learner Signature

Date

Facilitators' Signature

Date

ASSESSMENT FEEDBACK REPORT

FACILITATOR FEEDBACK & REMARKS

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ASSESSMENT JUDGEMENT

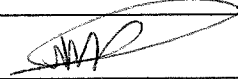
Learner's Total Mark: 100	Requirements met ☒	Requirements not met ☐
Action/s required:		
By when:		

LEARNER FEEDBACK & COMMENTS

Very good Sir Malinga.

DECLARATION BY THE FACILITATOR

I (Facilitator) hereby certify that I have examined the learner workbook and I am satisfied with the evidence provided by the learner.
--

DECLARATION BY LEARNER			
I, <u>Xolani Collen Malinga</u>declare that I am satisfied that the feedback given to me by the Facilitator was relevant, sufficient and done in a constructive manner. I accept the assessment judgment and have no further questions relating to this particular assessment event.			
			
Learner	Date	Facilitator	Date

DECLARATION BY THE ASSESSOR			
I (Assessor) hereby certify that I have examined the learner workbook and I am satisfied with the Facilitator Judgment of this assessment.			
Assessor	Date	Moderator	Date

Application for Assessment

First Attempt ☐ Second Attempt ☐ Third Attempt ☐ RPL ☐

Observer/Witness required: Yes/No

Name: _____

Contact Number: _____

Interpreter required: Yes/No

Name: _____

Contact Number: _____

Was Employer informed of request for Assessment: Yes/No

Type of Assessment: Theoretical

Practical

Recommendation for Assessment

The applicant has completed his/her training, has prepared for the assessment and is now ready to be assessed.

The ETDP Practitioner believe that the candidate is ready for the assessment because he has completed the relevant unit standard.

Notification of Assessment

Assessment will take place as follows:

	Theory	Practical
Date		
Time	9H00	9H00
Venue	Employer premises	Employer premises
Instruments	Written assignment	Practical Assessment

Action required from candidate:

Candidate need to prepare him/herself to either write the Questionnaire or conduct Practical Assessment in order to achieve competency towards the relevant unit standard.

ETDP PRACTITIONER

Assessment Planning

	<u>Theory</u>	<u>Practical</u>
Date of assessment:		
When would be the best time for the Assessment?	9H00	9H00

Have you received an application to assess the candidate? Yes

What assessment methods do you consider using? (observation, assignment, presentation, etc.) Explain method.

1. Written Assignment : Printed Questionnaire

2. Observation : As per Practical Assessment Guide

3. Interview : List of Questions as per Practical Assessment Guide after Practical Observation

What resources would you require to conduct the assessment?

1. Assessor toolkit
2. Theoretical assignment
3. Practical assessment
4. Learner toolkit

How would you sequence your activities? Explain

1. Record preparation process
2. Meet with group/candidate and explain assessment procedure – how you are going to conduct the assessment
3. Learner will write summative assessments
4. Conduct observation in workplace where applicable
5. Complete learner toolkit
6. Complete assessor toolkit

PRACTICALS

